

Heavi Energy — Precision Metric Framework

Status: Defined, pending design-partner pilot data · June 2026

1. Why precision, and why it is not yet measured

Heavi's published validation measures **recall**:

Of the real solar installations that developers actually built, what fraction does the tool score High?

Across 10 states, the answer is **71% among greenfield-eligible installations**, with positive discrimination versus random rural land in all 10 states. Recall answers "does the tool recognize good sites?" — and the validation says yes.

But recall is the wrong question for a developer deciding whether to rely on the tool. The developer's question is **precision**:

Of the parcels the tool scores High, what fraction would I actually want to develop?

Recall can be measured against public ground truth (EIA Form 860). **Precision cannot** — there is no public dataset of "parcels a developer reviewed and rejected." Measuring precision requires a domain expert reviewing Heavi's output and judging each High-scored parcel. That expert is the design partner. This document defines the precision metric **now** so it is ready to measure the moment pilots begin, with no methodology debate mid-pilot.

2. Definition

[illegible]

A High-scored parcel **passes developer review** if the design partner's site-development team classifies it as either:

- **Would pursue** — the parcel is genuinely suitable for development consideration; the team would advance it in their own pipeline.
- **Already known** — the parcel is already in the developer's pipeline. This *validates* the tool: it independently surfaced a site the experts had already chosen.

A High-scored parcel **fails developer review** if classified as:

- **Would not pursue** — the parcel has a real flaw the tool did not catch. Examples: wrong zoning, a terrain problem invisible at 10 m DEM resolution, known community opposition, a grid constraint not reflected in queue data, land already optioned by a competitor, or a parcel-specific title/access issue.

A **false positive** is a "would not pursue." Precision is the complement of the false-positive rate among High-scored parcels.

3. Why "already known" counts as a pass

A developer's first reaction to a tool surfacing a site they already hold is "it told me nothing new." The opposite is true: **independent rediscovery of an expert-selected site is the strongest possible confirmation that the scoring tracks real-world developer judgment.** A tool that only ever surfaced novel sites — and never the ones experts had already chosen — would be the suspicious one. Counting "already known" as a pass measures whether the tool *agrees with experts*, which is exactly what a screening tool should do.

We report both figures so a partner can see the split:

- **Precision (agreement):** (would-pursue + already-known) / total High
- **Net-new precision:** would-pursue / total High — the rate of *novel* good sites

4. Measurement Protocol

A single pilot run, repeatable per partner:

1. **Partner provides 50 candidate parcels** from their own screening pipeline (coordinates or addresses; any stage of their funnel).
2. **Heavi scores all 50** through the production solar-suitability platform, with full per-criterion breakdown, confidence, and interconnection context.
3. **Heavi identifies the top 15 as High** (rank by score; in practice the High band is ≥ 0.70 , but we fix $N = 15$ so precision is measured on a consistent denominator).
4. **The partner's team reviews the 15 High-scored parcels** against their own development criteria — the same review they would give any candidate site.
5. **The partner classifies each** as would-pursue / would-not-pursue / already-known.
6. **Compute precision** = (would-pursue + already-known) / 15.

Every "would not pursue" is logged with a **reason code** (zoning, terrain, grid, community, land-control, other). These reasons are the most valuable output of the pilot: they tell us which real-world constraint the tool should ingest next.

5. Target and interpretation

Target: $\geq 70\%$ precision — at least **10 of 15** High-scored parcels pass developer review.

Precision	Interpretation
$\geq 80\%$	Strong — the High band is reliable enough to drive real prioritization
70–79%	Target met — useful screening; review the failure reasons for the next data source
50–69%	Marginal — the tool narrows the field but the High band needs another constraint layer
$< 50\%$	The High band is not yet trustworthy for this partner's geography/segment; investigate

A precision below target is **not a failure of the pilot** — it is a prioritized backlog. If three of five "would not pursue" parcels failed on zoning, the next engineering investment is a parcel-level zoning layer, and the pilot has paid for itself by saying so.

6. Relationship to the recall validation

The two metrics are complementary and should always be reported together:

	Question	Ground truth	Status
Recall	Does the tool recognize good sites?	EIA Form 860 (public)	Measured: 71% greenfield-eligible
Precision	Are the tool's good sites actually good?	Design-partner expert review	Defined here; measured in pilots

A tool can have high recall and low precision (it finds the good sites but also flags many bad ones) or the reverse. A screening tool that saves a developer time needs **both**: high recall so it does not miss opportunities, and high precision so its shortlist is worth reviewing. The recall result is in hand; this framework closes the loop on precision.

7. Coverage (a secondary pilot metric)

Alongside precision, pilots measure **coverage**:

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`` Developer-selected parcels that Heavi also scores High Coverage =
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total developer-selected
parcels ``

If the developer independently advanced 20 of their 50 parcels, coverage asks how many of those 20 Heavi also placed in its High band. **Target: ≥80% coverage.** Coverage is recall measured on the partner's own ground truth rather than EIA's, and it is the metric most likely to convince a skeptical buyer, because it is *their* sites.

Heavi Energy. This framework is published in advance of pilot data so that precision is measured against a fixed, agreed definition. Results will be reported per partner, with failure-reason codes, as pilots complete.